# Environmental Product Declaration

EPD of multiple products, based on representative product.
In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for, and conformity with ISO 21930:2017:

## *PC strand*

*Diameter from 9.3 mm to 21.6 mm, listed on page 5, from*

## *Silvery Dragon Prestressed Materials Co. Ltd. Tianjin*

| | |
|---|---|
| Programme: | The International EPD® System, www.environdec.com |
| Programme operator: | EPD International AB |
| EPD registration number: | EPD-IES-0015453 |
| Publication date: | 2024-07-09 |
| Valid until: | 2029-07-08 |

*An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com*

# General information

## Programme information

| | |
|---|---|
| **Programme:** | The International EPD® System |
| **Address:** | EPD International AB<br>Box 210 60<br>SE-100 31 Stockholm<br>Sweden |
| **Website:** | www.environdec.com |
| **E-mail:** | info@environdec.com |

| **Accountabilities for PCR, LCA and independent, third-party verification** |
|---|
| **Product Category Rules (PCR)** |
| CEN standard EN 15804 serves as the Core Product Category Rules (PCR) |
| Product Category Rules (PCR): PCR 2019:14 Construction products Version 1.3.4 [valid until: 2025-06-20] |
| PCR review was conducted by: The Technical Committee of the International EPD System. The review panel may be contacted via info@environdec.com. |
| **Life Cycle Assessment (LCA)** |
| LCA accountability: Sijia Yang from IVL Swedish Environmental Research Institute and Shuhan Huang from IVL Environmental Technologies Company Ltd. |
| **Third-party verification** |
| Independent third-party verification of the declaration and data, according to ISO 14025:2006, via:<br><br>☒ EPD verification by individual verifier<br><br>Third-party verifier: Matthew Fishwick from Fishwick Environmental Ltd.<br><br>Approved by: The International EPD® System |
| Procedure for follow-up of data during EPD validity involves third party verifier:<br>☐ Yes ☒ No<br><br>[Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier] |

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs;

cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

The results of the end-of-life stage (module C) should be considered when using the results of the production stage (modules A1-A3).

## Company information

Owner of the EPD: Silvery Dragon Prestressed Materials Co. Ltd. Tianjin

Contact: Samantha Sun (sales@yinlong.com)

Website: www.yinlong.com

Description of the organisation:
Silvery Dragon Co., Ltd., originated from Hejian, the old city with a history of thousands of years. For forty-five years, adhering to innovation, ingenuity, and perseverance, focusing on high-performance prestressed materials, rail transit concrete products, high-tech equipment, and information-based industry, offers service for multiple fields domestic and abroad, including rail transit, highways, railways, water conservancy, new energy, construction and so on.

Product-related or management system-related certifications:
**For products:**
AS/NZS 4672:2007, nHRN EN 10138-3 (prEN 10138-3:2000), nHRN EN 10138-1 (prEN 10138-1:2000), BS 5896:2012, MS 1138 : PART 4 : 2007, ASTM A416/A416M-18, ASTM A 416/A 416M:17a, Israel Standard No. 1735 Part 4 Edition November 2022, JIS G 3536, and prEN 10138-3:2009

**For management system:**
ISO 9001 Quality Management System Certification, GB/T 23331-2020/SO 50001:2018 Energy Management Systems-Requirement with Guidance for Use and the certification and accreditation industry standard: RB/T 103-2013 Energy Management System-Certificate Requirement for Iron and Steel Industry

Name and location of production site:
Baozelong Metal Materials Co., Ltd. Hejian: No.1 North Jianshe Street, Economic and Technological Development Area, Hejian City, Hebei Province, China

## Product information

Product name:
PC strand (pre-stressed concrete steel strand)

Product description:
PC strand is a prestressed material with a rounded or regularly deformed surface, made of 3 or 7 twisted steel wires. Its service life depends on the service life of the prestressed concrete components. This LCA analysis covers the diameter of PC strand from 9.3 mm to 21.6 mm. The figure of the PC strand is shown as below.

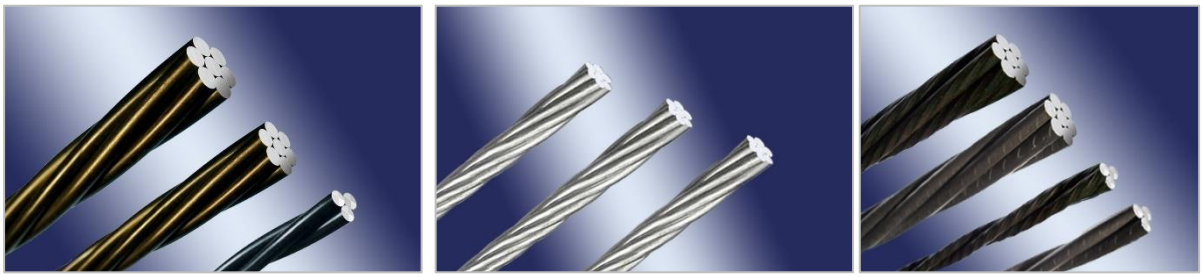


Figure 1. Picture of PC strand products

This EPD is for multiple products, based on representative product. In this study, the representative product of the product family is 7-wire 15.2 mm diameter plain PC strand. The selection criteria of the representative product is the yield rate in the studied time. The representative product has the highest yield rate of the product family. The family of products are all made of steel wire rod, some of which are then galvanised with a layer of zinc on the surface of the product. The functionality of the products is basically the same, but they are made in different thicknesses in order to meet the different requirements of the project. The list and performance of the products in this family is listed in the table below, provided by Silvery Dragon.

Table 1. Product list and performance of the PC strand product

| Appearance | Nominal Diameter (mm) | Tensile Strength (MPa) | Relaxation (1000h) | Standards |
|---|---|---|---|---|
| 7 wires | 8.0, 9.3, 9.53, 11.1, 12.5, 12.7, 12.9, 15.2, 15.7, 17.8, 21.6 | 1770, 1860, 2000 | Low relaxation ≤2.5% | ASTMA416, BS5896, EN10138-3, AS/NZS4672, GB/T5224, KS7002, ISO6934-4, SS213620, JIS G3536, UNE36094, ABNT NBR7483, NEN3868 |
| 3 wires | 4.8, 5.2, 5.8, 6.2, 6.5, 7.5, 7.6, 8.6, 9.1 | 1725, 1860, 1960 | | ASTMA910, GB/T5224, TISG3536, EN10138-3, AS/NZS4672 |

UN CPC code:
412

Geographical scope:
Modules A1-A3 represent production of the products in China. Module A4 is not declared. Module A5 is assumed the waste package treatment in global. Module B is not declared. Module C and D represent the end-of-life treatment with benefits out of the system boundary of the products in in global.

## LCA information

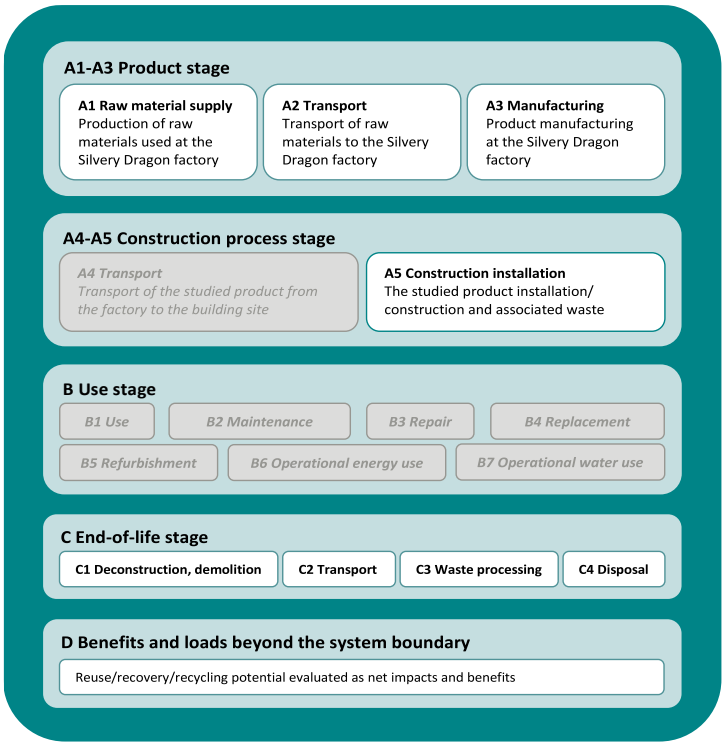
Declared unit:
1 tonne of steel strand (PC strand).

Time representativeness:
2022 November to 2023 October.

Database(s) and LCA software used:
Managed LCA Content 2023.1 Databases (2023.1 Edition) and Ecoinvent 3.9.1, LCA for Expert (Gabi).

Description of system boundaries:
The scope of the EPD generated corresponds to “cradle to gate with options” which serves type (b) EPD, assessing the potential environmental impacts associated with the studied product. The information module included in the study is A1-A3, A5, C1-C4, and D. Module A4 and B1-B7 are not declared in the EPD.

System diagram:




Manufacturing processes:

As the manufacturing process is important to understand the whole studied system, the description of the main manufacturing processes for producing the studied product is explained as below. A flow chart of product manufacturing is shown below.

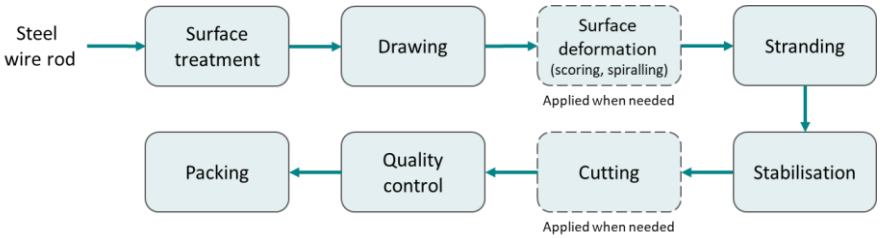



Figure 2. The main production process of studied product.

PC strand is made from steel wire rod. The first step is surface treatment, all input steel wire rods need be treated by sulphuric acid and phosphate accelerator, boriding agent may be used when needed. Then the treated steel wire rod is drawn through drawing machine with drawing powder. For some of PC strand products, the steel wire rod after drawing is sent to surface deformation process for scoring or spiralling if needed. Then, for PC strand products, three or seven steel wire rods are stranded together for forming the strand. After this, all semi-finished products are sent to stabilisation process. The cutting process could be applied when needed. The final products which finished the quality control process, are sent the packing line to get the package.

The GWP-GHG of electricity mix for Hebei province in this study is 0.825 kg $CO_2$ eq./kWh. The electricity mix could be found in Table 2.

Table 2. Electricity structure of the Hebei province and the dataset chosen for it

| Electricity generation sources | Dataset used in the model | GWP-GHG ($CO_2$ eq./kWh) | Percentage in Hebei province |
|---|---|---|---|
| **Electricity from fossil fuel(1)** | CN: Electricity from hard coal Sphera | 1.11 | 73.52% |
| **Electricity from hydro power** | CN: Electricity from hydro powe | 0.00745 | 0.78% |
| **Electricity from nuclear power** | CN: Electricity from nuclear power Sphera | 0.00446 | 0.00% |
| **Electricity from wind power** | CN: Electricity from wind power Sphera | 0.0171 | 16.62% |
| **Electricity from photovoltaic** | CN: Electricity from photovoltaic Sphera | 0.0288 | 9.08% |

(1) In the 2022 China Electricity Yearbook, the percentage of electricity from fossil fuel for each province is not specified. A brief description of electricity from fossil fuel for the whole country is given in the yearbook, i.e. it covers coal, gas, oil, biomass, and a small amount of unidentified sources for generating electricity. Based on the information in the yearbook, i.e., electricity from hard coal accounts for more than 80% of the thermal power generation types, and considering that China is a country where coal-fired power generation is the main source of thermal power generation, the LCA practitioner (IVL) decide to use electricity from hard coal as 100% of the dataset selection for electricity from fossil fuel in this study for modelling.

## More information:

Scenarios and additional technical information:

- The products are manufactured in China, and the waste treatment of the waste generated during the manufacturing process was included in the system boundary.
- The products are sent and used globally. Regarding A5 module, since lacking supporting information, the environmental impact during the construction stage of the product is not considered. However, since the package contains biogenic carbon, depending on the recommendation of the PCR, it is assumed that the packaging becomes to waste in this stage and is treated to end-of-life stage. So the package treatment with the biogenic carbon dioxide is considered in this stage.
- The modelling of the C1 module is assumed that the consumption of additional materials and energy used in the deconstruction stage for installation is zero since lacking information. After C1, it is assumed the 15% of the waste product would be sent to C4 for disposal, and 85% for recycling according to the worldsteel data (the link is listed in the reference list). For the C2 module, the assumption has been made that the all waste product would be transported for 100 km by truck. For the waste processing C3 module, disposal C4 module, and D module (benefits and loads beyond the system boundary), the generic data has been applied. In detail, it is assumed the waste product would be processed in C3 with a 3.01% mass loss depending on Sphera.
- About module D calculation, since the products have post-consumer material (only steel scrap) as the raw material, all steel scrap is excluded from the calculation of the benefit to avoid the double counting of the benefit. Meanwhile, this study does not consider the co-product allocation in this project, so there is no co-product allocation issue for the module D calculation. Besides, the mass loss during module D is assumed to be zero, and we used the datasets and the plan directly from the Sphera database to calculate module D.

Allocation:
Allocation rules for multifunctional products and multiproduct processes are mentioned in PCR. Allocation rules for co-products are mentioned in the PCR. In this study, the studied product has co-products produced together. However, after considering the economic allocation, the studied product needs to afford over 99.29% of the environmental burden. After discussing with Silvery Dragon, no co-product allocation has been applied in this study i.e. all burdens are allocated to the final studied product. For the allocation of waste, module D calculation, and scarp, this study strictly follows the PCR.

Cut-off rules:
The cut-off criteria established by the PCR is that data for elementary flows to and from the product system contributing to a minimum of 95% of the declared environmental impacts shall be included (not including processes that are explicitly outside the system boundary). No cut-off for the studied product has been applied in the study.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

| | Product stage | | | Construction process stage | | Use stage | | | | | | | End of life stage | | | | Resource recovery stage |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Raw material supply | Transport | Manufacturing | Transport | Construction installation | Use | Maintenance | Repair | Replacement | Refurbishment | Operational energy use | Operational water use | De-construction demolition | Transport | Waste processing | Disposal | Reuse-Recovery-Recycling-potential |
| **Module** | **A1** | **A2** | **A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| Modules declared | X | X | X | ND | X | ND | ND | ND | ND | ND | ND | ND | X | X | X | X | X |
| Geography | CN | CN | CN | - | GLO | - | - | - | - | - | - | - | GLO | GLO | GLO | GLO | GLO |
| Specific data used | 12.29% | | | | | - | - | - | - | - | - | - | - | - | - | - | - |
| Variation – products | <10% | | | | | - | - | - | - | - | - | - | - | - | - | - | - |
| Variation – sites | 0% | | | | | - | - | - | - | - | - | - | - | - | - | - | - |

(1) Modules included in the EPD (X) and the modules not declared (ND).

# Content information

The representative product of this EPD is 7-wire 15.2 mm diameter plain PC strand and the content of the product with its package is shown below:

Representative product, showing in per declared unit:

| Product components | Weight, kg | Post-consumer material, weight-% of total product | Biogenic material, kg C/product |
|---|---|---|---|
| Steel wire rod | 1 000.00 | 20.00% | 0.00 |
| TOTAL | 1 000.00 | 20.00% | 0.00 |
| **Packaging materials** | **Weight, kg** | **Weight-% (versus the product)** | **Weight biogenic carbon, kg C/product** |
| Packing steel straps | 2.80 | 0.28% | 0.00 |
| Product label | 0.50 | 0.05% | 0.23 |
| Anti-rust paper with woven | 1.10 | 0.10% | 0.05 |
| Wooden block | 2.00 | 0.20% | 1.18 |
| TOTAL | 6.40 | 0.64% | 1.46 |

The content declaration of the product family is shown below.

The raw material composition range of the product family (corresponding to one declared unit):

| Product composition | Percentage range of the product family, weight-% of whole product |
|---|---|
| Steel wire rod | 95.69-100.00% |
| Zinc | 0.00%-2.39% |
| Aluminium-zinc alloy | 0.00%-1.91% |
| TOTAL | 100.00% |
| **Packaging Material composition** | **Packaging weight, kg** |
| Packing steel straps | 2.80 |
| Product label | 0.50 |
| Anti-rust paper with woven | 1.10 |
| Wooden block | 2.00 |
| TOTAL | 6.40 |

At the time of data collection, no substance included in the Candidate List of Substances of Very High Concern (SVHC) for authorization under the REACH Regulations is present in the products covered by this LCA and EPD either above the threshold for registration with the European Chemicals Agency or above 0.1% (wt/wt).

Depending on the PCR, in general, the production and end-of-life processes of infrastructure or capital goods used in the product system should be excluded, unless there is evidence that they are relevant in terms of their environmental impact, or when a generic LCI dataset includes infrastructure/capital goods, and it is not possible, within reasonable effort, to subtract the data on infrastructure/capital goods from this dataset (directly citation from section 4.3.2 of PCR 1.3.4). In this study, the infrastructure and capital goods are not included in the LCA analysis since they are used plenty of times for several years for the product manufacturing. According to the PCR, it should be excluded.

All results in this LCA analysis are calculated by the EN 15804+A2. The “EN 15804 reference package” is calculated based on EF 3.1.

# Results of the environmental performance indicators

The declared unit in the study is 1 tonne of steel strand (PC strand).

The estimated impact results are only relative statements which do not indicate the end points of the impact categories, exceeding threshold values, safety margins or risks.

## Mandatory impact category indicators according to EN 15804

| Results 1 tonne of steel strand (PC strand) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A5** | **C1** | **C2** | **C3** | **C4** | **D** |
| GWP-fossil | kg $CO_2$ eq. | 2.49E+03 | 1.57E+00 | 0.00E+00 | 9.58E+01 | 2.17E+00 | 2.64E+00 | -1.19E+03 |
| GWP-biogenic | kg $CO_2$ eq. | -1.36E+01 | 7.24E+00 | 0.00E+00 | 4.75E-02 | -2.02E-02 | 1.36E+01 | 6.81E+00 |
| GWP-luluc | kg $CO_2$ eq. | 1.24E+00 | 1.06E-04 | 0.00E+00 | 2.62E-03 | 1.64E-02 | 8.21E-03 | -1.52E-01 |
| GWP-total | kg $CO_2$ eq. | 2.48E+03 | 8.81E+00 | 0.00E+00 | 9.59E+01 | 2.17E+00 | 1.62E+01 | -1.18E+03 |
| ODP | kg CFC 11 eq. | 4.14E-05 | 8.14E-13 | 0.00E+00 | 5.74E-12 | 3.64E-12 | 6.72E-12 | 1.54E-09 |
| AP | mol $H^+$ eq. | 9.58E+00 | 2.04E-03 | 0.00E+00 | 1.34E-01 | 1.13E-02 | 1.88E-02 | -2.80E+00 |
| EP-freshwater | kg P eq. | 9.42E-01 | 3.64E-07 | 0.00E+00 | 1.22E-05 | 7.40E-06 | 5.32E-06 | -2.67E-04 |
| EP-marine | kg N eq. | 2.19E+00 | 7.42E-04 | 0.00E+00 | 4.68E-02 | 5.21E-03 | 4.85E-03 | -4.50E-01 |
| EP-terrestrial | mol N eq. | 2.30E+01 | 9.30E-03 | 0.00E+00 | 5.28E-01 | 5.75E-02 | 5.33E-02 | -4.03E+00 |
| POCP | kg NMVOC eq. | 1.11E+01 | 1.95E-03 | 0.00E+00 | 1.19E-01 | 1.41E-02 | 1.46E-02 | -1.83E+00 |
| ADP-minerals& metals* | kg Sb eq. | 3.09E-03 | 9.25E-10 | 0.00E+00 | 6.34E-07 | 2.33E-06 | 1.22E-07 | -6.48E-03 |
| ADP-fossil* | MJ | 3.14E+04 | 2.39E+00 | 0.00E+00 | 1.37E+03 | 4.28E+01 | 3.52E+01 | -1.14E+04 |
| WDP* | $m^3$ | 7.19E+02 | 5.68E-01 | 0.00E+00 | 4.28E-01 | 4.23E-01 | 2.90E-01 | -7.73E+01 |
| Acronyms | GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption | | | | | | | |

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

## Additional mandatory and voluntary impact category indicators

| Results 1 tonne of steel strand (PC strand) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A5** | **C1** | **C2** | **C3** | **C4** | **D** |
| GWP-GHG[1] | kg $CO_2$ eq. | 2.37E+03 | 1.56E+00 | 0.00E+00 | 9.47E+01 | 2.14E+00 | 2.61E+00 | -1.14E+03 |

## Resource use indicators

| Results 1 tonne of steel strand (PC strand) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A5** | **C1** | **C2** | **C3** | **C4** | **D** |
| PERE | MJ | 2.51E+03 | 4.95E-01 | 0.00E+00 | 7.53E+00 | 3.98E+00 | 5.74E+00 | 4.50E+02 |
| PERM | MJ | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| PERT | MJ | 2.51E+03 | 4.95E-01 | 0.00E+00 | 7.53E+00 | 3.98E+00 | 5.74E+00 | 4.50E+02 |
| PENRE | MJ | 3.14E+04 | 2.39E+00 | 0.00E+00 | 1.37E+03 | 4.29E+01 | 3.52E+01 | -1.14E+04 |
| PENRM | MJ | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| PENRT | MJ | 3.14E+04 | 2.39E+00 | 0.00E+00 | 1.37E+03 | 4.29E+01 | 3.52E+01 | -1.14E+04 |
| SM | kg | 2.00E+02 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| RSF | MJ | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| NRSF | MJ | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| FW | $m^3$ | 1.68E+01 | 1.33E-02 | 0.00E+00 | 1.15E-02 | 1.22E-02 | 8.89E-03 | -1.16E+02 |
| Acronyms | PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water | | | | | | | |

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic $CO_2$ is set to zero.

### Waste indicators

| Results 1 tonne of steel strand (PC strand) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A5** | **C1** | **C2** | **C3** | **C4** | **D** |
| Hazardous waste disposed | kg | -1.68E-07 | 8.21E-11 | 0.00E+00 | 3.40E-10 | -1.11E-10 | 7.67E-10 | -8.52E-05 |
| Non-hazardous waste disposed | kg | 2.67E+00 | 3.22E+00 | 0.00E+00 | 5.19E-02 | 1.13E-02 | 1.76E+02 | 1.38E+02 |
| Radioactive waste disposed | kg | 3.97E-03 | 9.40E-05 | 0.00E+00 | 4.94E-04 | 5.75E-04 | 4.01E-04 | 1.25E-03 |

### Output flow indicators

| Results 1 tonne of steel strand (PC strand) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A5** | **C1** | **C2** | **C3** | **C4** | **D** |
| Components for re-use | kg | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| Material for recycling | kg | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| Materials for energy recovery | kg | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| Exported energy, electricity | MJ | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |
| Exported energy, thermal | MJ | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 | 0.00E+00 |

## Additional information

### Hot-spot analysis

In this LCA study, the bulk of GWP-GHG impact is in A1-A3 and 87% comes from raw materials and 12% from manufacturing, and the rest is due to the transportation.

# References

Emami, N., Heinonen, J., Marteinsson, B., Säynäjoki, A., Junnonen, J.-M., Laine, J., & Junnila, S. (2019). A Life Cycle Assessment of Two Residential Buildings Using Two Different LCA Database-Software Combinations: Recognizing Uniformities and Inconsistencies. Buildings, 9(1), 20. https://doi.org/10.3390/buildings9010020

Global Guidance Principles for LCA databases. A basis for greener processes and products, 'Shonan Guidance Principles', 2011; ISBN: 978-92-807-3174-3.

GPI (2021), General Programme Instructions for the International EPD System version 4.0.

ISO (2006a). ISO 14025:2006, Environmental labels and declarations – Type III environmental declarations – Principles and procedures.

ISO (2006b). ISO 14040:2006, Environmental management – Life cycle assessment – Principles and framework.

ISO (2006c). ISO 14044: 2006, Environmental management – Life cycle assessment – Requirements and guidelines.

Joint Research Centre of European Commission (2023). Updated characterisation and normalisation factors for the Environmental Footprint 3.1 method.

Kalverkamp, M., Helmers, E., & Pehlken, A. (2020). Impacts of life cycle inventory databases on life cycle assessments: A review by means of a drivetrain case study. Journal of Cleaner Production, 269, 121329. https://doi.org/10.1016/j.jclepro.2020.121329

LCA database published by the ecoinvent association originally known as the ecoinvent Centre, the Swiss Centre for Life Cycle Inventories. Since June 2013 ecoinvent is a not-for-profit association founded by institutes of the ETH Domain and the Swiss Federal Offices. The cut-off version 3.9.1 was used.

LCA for Expert software. The former name of the software is Gabi. The software version 10.7, CUP 2023.1 version is used.

Managed LCA Content 2023.1 Databases (Gabi database) provided by Sphera.

Pauer, E., Wohner, B., & Tacker, M. (2020). The Influence of Database Selection on Environmental Impact Results. Life Cycle Assessment of Packaging Using GaBi, Ecoinvent 3.6, and the Environmental Footprint Database. Sustainability, 12(23), 9948. https://doi.org/10.3390/su12239948

PCR Construction, PCR 2019:14, Version 1.3.4, Valid until 2025-06-20

Silvery Dragon products promotional documents (2023).

Silvery Dragon technical files (2023).

Sphera. The provider of the LCA for Expert software and database.

The International EPD system. https://www.environdec.com

Worldsteel (2018). Steel – the surprising recycling champion. https://worldsteel.org/media-centre/blog/2018/steel-surprising-recycling-champion/

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www.environdec.com